

FreeBSD relies on `umass` driver that uses a SCSI system for accessing USB devices. As the system can see any USB device as a SCSI device, if the USB device is, say, an external CD burner, do not add `device atapicam` in the custom kernel configuration file.

Device configuration

To check your USB configuration, you will obviously need to plug in a USB device. You can use `dmesg` to see the messages displayed when you plug in the device. This might look like:

- `umass0: <STECH Simple Drive, class 0/0, rev 2.00/1.04, addr 3> on usb0`
- `umass0: SCSI over Bulk-Only; quirks = 0x0100`
- `umass0:4:0:-1: Attached to scbus4`
- `da0 at umass-sim0 bus 0 scbus4 target 0 lun 0`
- `da0: <STECH Simple Drive 1.04> Fixed Direct Access SCSI-4 device`
- `da0: Serial Number WD-WXE508CAN263`
- `da0: 40.000MB/s transfers`
- `da0: 152627MB (312581808 512 byte sectors: 255H 63S/T 19457C)`
- `da0: quirks=0x2<NO_6_BYTE>`

The brand, speed, device node (da0), and size can vary as per the device. As the USB is perceived as a SCSI, `cam control` can be used to list the attached USB storage devices to the system.

- `# camcontrol devlist`
- `<STECH Simple Drive 1.04> at scbus4 target 0 lun 0 (pass3,da0)`

Or, `usbconfig` command can be used, which would look like:

- `# usbconfig`
- `ugen0.3: <Simple Drive STECH> at usb0, cfg=0 md=HOST spd=HIGH (480Mbps) pwr=ON (2mA)`

Note: Giving access to the untrusted users of the devices for mounting arbitrary media by allowing the command `vfs.usermount` (as mentioned below), is not considered secure. Most file systems are not developed to prevent malicious devices or attacks.

- `[localrules=5]`
- `add path 'da*' mode 0660 group operator`

To make USB devices mountable by just any general user, you can make all the users of the devices members of the main operator group with the help of `pw`. You can also make sure that the operator is capable of reading